FIITJEE

ALL INDIA TEST SERIES

<u>FULL TEST – XII</u>

JEE (Main)-2022

<u>TEST DATE: 22-05-2022</u>

Time Allotted: 3 Hours **Maximum Marks: 300**

<u>General Instructions:</u>

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Physics

PART – A

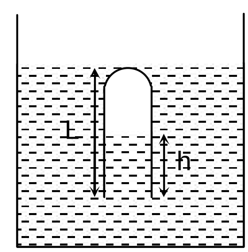
SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. If the area under the I-H hysterics loop and B-H hysterics loop are denoted by A_1 and A_2 then
   (A) $A_2 = \mu_0 A_1$
   (B) $A_2 = A_1$
   (C) $A_1 = \mu_0 A_2$
   (D) $A_2 = \mu_0^2 A_1$

2. There is a set of 4 tunning forks, one with lowest frequency vibrating at 552 Hz. By using two forks at a time the beat frequencies heard are 1, 2, 3, 5, 7, 8. then the possible frequencies of other three forks are
   (A) 553 Hz, 554 Hz, 560 Hz
   (B) 553 Hz, 555 Hz, 560 Hz
   (C) 553 Hz, 556 Hz, 558 Hz
   (D) 551 Hz, 554 Hz, 560 Hz

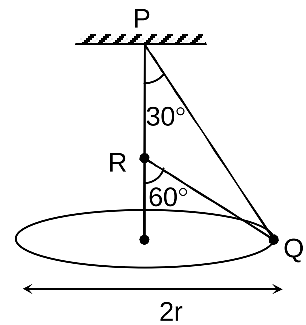
3. A tube of length L is open at one end. It is lowered into a tank of liquid as shown in the figure with open end downward. The closed end just touches the surface. The liquid is pushed into the tube to a height h. If atmospheric pressure is L of liquid, then choose the correct one.



   (A) $(2L + h)(L - h) = L^2$
   (B) $(2L + h)(L + h) = L^2$
   (C) $(2L - h)(L + h) = L^2$
   (D) $(2L - h)(L - h) = L^2$

4. The centres of two thin coaxial conducting loops of radius r each are separated by distance d (d >> r). Then, the mutual inductance of the system is:
   (A) $\dfrac{\mu_0 \pi r^2}{d}$
   (B) $\dfrac{\mu_0 \pi r^4}{d^3}$
   (C) $\dfrac{\mu_0 \pi r^2}{2d}$
   (D) $\dfrac{\mu_0 \pi r^4}{2d^3}$

5. A smooth wire PQR passes through a bead Q which rotates with a constant angular speed in the horizontal circle of radius r, as shown in figure. The angular speed is



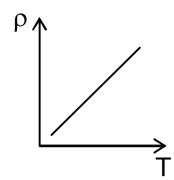
(A) $\sqrt{\frac{g}{r}}$

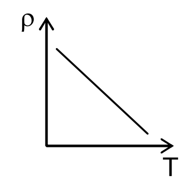
(B) $\sqrt{\frac{g}{2r}}$

(C) $\sqrt{\frac{2g}{r}}$

(D) $\sqrt{\frac{g}{3r}}$

6. The temperature (T) dependence of resistivity (ρ) of semiconductor is represented by

(A)


(B)


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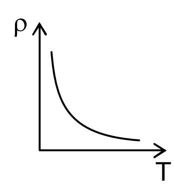

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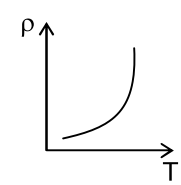
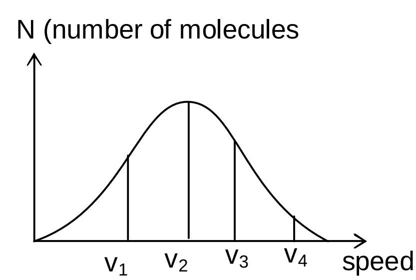


7. For the given Maxwell's speed distribution curve, if v_P = most probable speed, v_{rms} = r.m.s. speed and v_m = mean speed, then



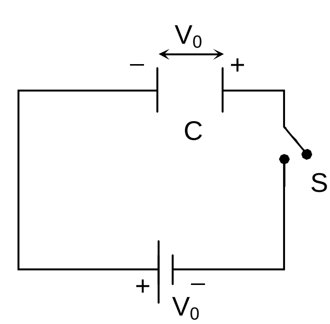
(A) $v_1 = v_{rms},\ v_2 = v_P,\ v_3 = v_m$

(B) $v_2 = v_P,\ v_3 = v_m,\ v_4 = v_{rms}$

(C) $v_2 = v_P,\ v_3 = v_{rms},\ v_4 = v_m$

(D) $v_2 = v_m,\ v_3 = v_P,\ v_4 = v_{rms}$

8. Find the amount of heat produced in the circuit after closing the switch (capacitor is initially charged with potential V_0 as polarity shown in the figure)



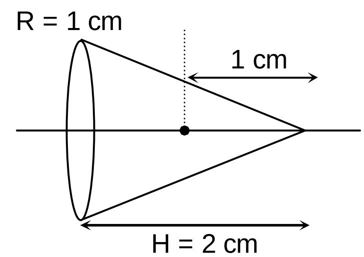
(A) zero

(B) $\frac{1}{2}CV_0^2$

(C) CV_0^2

(D) $2\,CV_0^2$

9. A positive charged particle is placed inside the hollow right circular cone on its axis at a distance of 1 cm from the apex as shown. Find the electric flux passing through the upper half curved surface of cone.



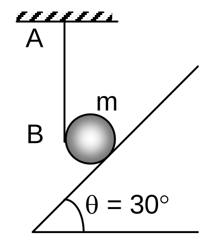
(A) $\frac{q}{2\varepsilon_0}(\sqrt{2}-1)$

(B) $\frac{q}{4\varepsilon_0}\left(1-\frac{1}{\sqrt{2}}\right)$

(C) $\frac{q}{2\varepsilon_0}(\sqrt{2}+1)$

(D) $\frac{q}{4\varepsilon_0}\left(1+\frac{1}{\sqrt{2}}\right)$

10. A solid sphere of mass m is held in equilibrium on a rough inclined surface with the help of an ideal string which is wrapped on it and one end is fixed such that part AB is vertical. Then the friction force acting on the system by the inclined surface is



(A) $\frac{mg}{2}$

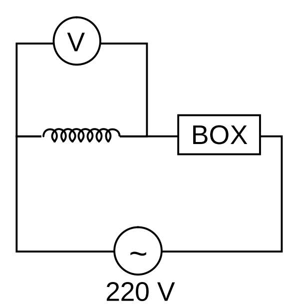
(B) $\frac{3mg}{4}$

(C) $\frac{mg}{3}$

(D) $\frac{2mg}{3}$

11. In the given circuit, if the reading of ideal voltmeter is 220 V, then the box may contain.

(A) only capacitor
(B) Series combination of capacitor and inductor
(C) Both (A) and (B)
(D) none of the above



12. A particle starts from rest and moves in a straight line with initial acceleration a_0. The acceleration reduces exponentially and becomes half of its initial value in t_0 seconds. The terminal velocity of the particle is

(A) $\frac{a_0 t_0}{\ell n2}$

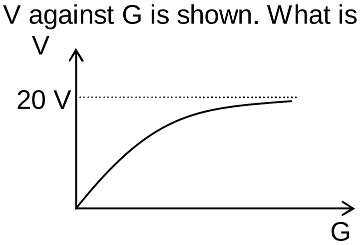
(B) $\frac{a_0 t_0}{2\ell n2}$

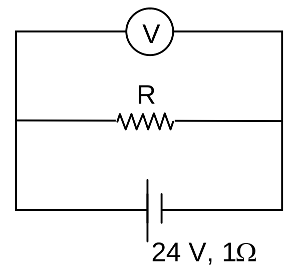
(C) $\frac{a_0 t_0}{3\ell n2}$

(D) $\frac{2a_0 t_0}{3\ell n2}$

13. According moseley's Law, the ratio of slopes of graph of $\sqrt{f}$ and Z for k_β and k_α is
(A) $\sqrt{\frac{32}{27}}$
(B) $\sqrt{\frac{27}{32}}$
(C) $\sqrt{\frac{33}{22}}$
(D) $\sqrt{\frac{22}{33}}$

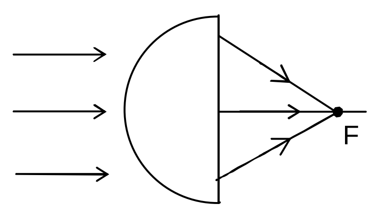
14. A cell of internal resistance 1Ω is connected across a resistor. A voltmeter having variable resistance G is used to measure potential difference across resistor. The plot of voltmeter reading V against G is shown. What is the value of R.



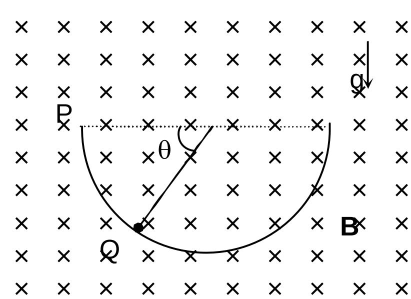


(A) 5Ω
(B) 4Ω
(C) 3Ω
(D) 1Ω

15. A paraxial beam is incident on a glass (n = 1.5) hemisphere of radius R = 6 cm in air as shown. The distance of point of convergence F from the plane surface is
(A) 5.4 cm
(B) 8 cm
(C) 12 cm
(D) 18 cm



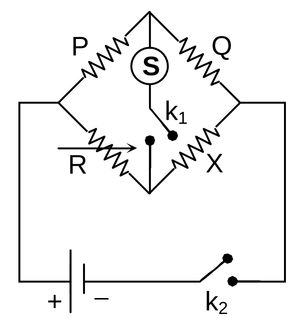
16. A charged sphere of mass m and charge –q starts sliding along the surface of a smooth hemispherical bowl from position P. The region has a transverse uniform magnetic field B. Normal force by the surface of bowl on the sphere at position Q.
(A) $mg\sin\theta + qB\sqrt{2gR\sin\theta}$
(B) $mg\sin\theta - qB\sqrt{2gR\sin\theta}$
(C) $3mg\sin\theta + qB\sqrt{2gR\sin\theta}$
(D) $3mg\sin\theta - qB\sqrt{2gR\sin\theta}$



17. The property of metals which allows them to be drawn readily into thin wires beyond the elastic limit without rupturing, is known as
(A) elastically
(B) Hardness
(C) Malleability
(D) Ductility

18. A large parallel plate capacitor, whose plates have an area of 1 m^2 and are separated from each other by 1 mm, is being charged at a rate of 25 V/s. If the dielectric between the plates has the dielectric constant 10, then the displacement current at this instant is
(A) 1.1 μA
(B) 11 μA
(C) 2.2 μA
(D) 0.22 μA

19. In a wheat-stone bridge experiment as shown in the figure.
(A) Key k_1 should be pressed first and then k_2
(B) Key k_2 should be pressed first and then k_1
(C) Any key can be pressed in any order
(D) Both keys should be pressed simultaneously.



20. A radioactive sample undergoes a series of decays according to the scheme
$A \xrightarrow{\alpha} A_1 \xrightarrow{\beta} A_2 \xrightarrow{\alpha} A_3 \xrightarrow{\gamma} A_4$
If the mass number and atomic number of A are 180 and 72 respectively, these numbers of A_4 are
(A) 172, 68
(B) 172, 69
(C) 172, 67
(D) 171, 69

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. The maximum vertical distance through which a full dressed astronaut can jump on the earth is 0.5 m. Estimate the maximum vertical distance (in m) through which he can jump on the moon. (Given $\rho_{moon} = \left(\frac{2}{3}\right)\rho_{earth}$, $R_{moon} = \left(\frac{1}{4}\right)R_{earth}$)

22. Two moles of helium are mixed with n moles of hydrogen. The root mean square speed of the gas molecules in the mixture is $\sqrt{2}$ times the speed of sound in mixture. The value of n is

23. The limiting angle of incidence of a ray that can be transmitted by an equilateral prism of refractive index $\sqrt{\frac{7}{3}}$ is $\frac{\pi}{4k}$ radian. Find k.

24. The ratio of heat required to increase the temperature of 2 moles of diatomic gas at constant pressure by 20°C to the heat required to increase the temperature of 3 moles of monoatomic gas at constant volume 14°C is

25. Radiation of wavelength $\frac{\lambda_0}{3}$ is incident on a photosensitive sphere of radius R. The charge developed on the sphere when electrons cease to be emitted will be $\left(\frac{6\pi\varepsilon_0 RChn}{e\lambda_0}\right)$. ($\lambda_0$ being the threshold wavelength). Find n.

26. A thin copper wire of length ℓ is heated from 20°C to 70°C, then increase in length is 1%. If a thin rectangular copper plate of length 3ℓ and breadth 2ℓ is heated through the same temperature change, then % change in its area will be

27. Starting from mean position a body oscillates simple harmonically with a period of 2s. Its kinetic energy will become 75% of the total energy after n sec for the first time. Find n.

28. An inductor coil stores 32 J of magnetic energy and dissipates it as heat at rate of 320 W when a current of 4A passed through it. The time constant of the coil is t_0 sec. Then find t_0

29. A faulty thermometer reads 1°C at freezing point of water and 99°C at boiling point of water. What is the reading of this faulty thermometer in °C when the correct thermometer reads 40°C.

30. In YDSE apparatus, d = 1mm, λ = 600 nm, D = 1m. Then find the minimum distance between two points on the screen (in mm) having 75% of the maximum intensity.

Chemistry PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Which of the following is correct for last electron filled in 'S' atom?

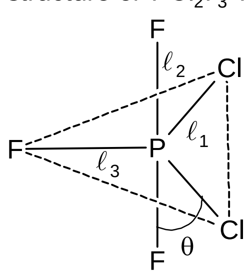
(A) $n = 3;\ \ell = 3;\ m = -2;\ s = +\frac{1}{2}$

(B) $n = 3;\ \ell = 1;\ m = -1;\ s = +\frac{1}{2}$

(C) $n = 3;\ \ell = 1;\ m = -1;\ s = -\frac{1}{2}$

(D) Both (B) and (C) are possible

32. Which of the following is correct about marked bond length and bond angles, in the following structure of PCl_2F_3 ?



(A) $\ell_2 > \ell_1 > \ell_3;\ \theta < 90^o$

(B) $\ell_1 > \ell_3 > \ell_2;\ \theta > 90^o$

(C) $\ell_1 > \ell_2 > \ell_3;\ \theta = 90^o$

(D) $\ell_1 > \ell_2 > \ell_3;\ \theta \simeq 120^o$

33. What is the concentration of $Cl_2(g)$, when 2 moles of $ClF_3(g)$ is allowed to decompose in a closed container of 5 litre at 304.5 K, with $K_P = 50\ atm^2$?

(A) 0.103 M
(B) 1.03 M
(C) 0.065 M
(D) 0.129 M

34. What is the solubility of AgBr in 0.1 M NH_3.

Given: $K_{SP}(AgBr) = 5 \times 10^{-13};\ K_f\left[Ag(NH_3)_2\right]^+ = 10^7$?

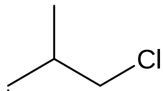
(A) 2.226×10^{-4} M

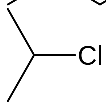
(B) 2.5×10^{-4} M

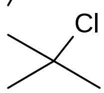
(C) 2.226×10^{-3} M

(D) 2.5×10^{-3} M

35. Which of the following undergo fastest S_N2 by NaBr/acetone?

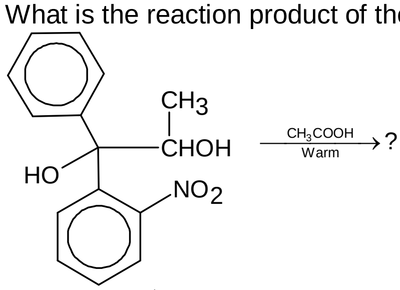
(A) 

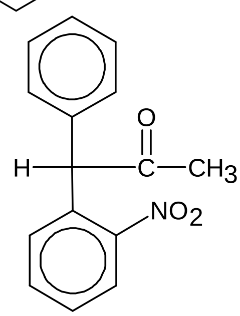
(B) 

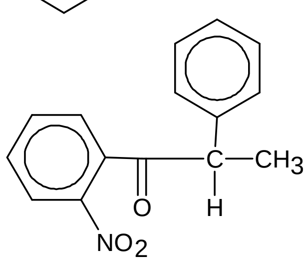
(C) 

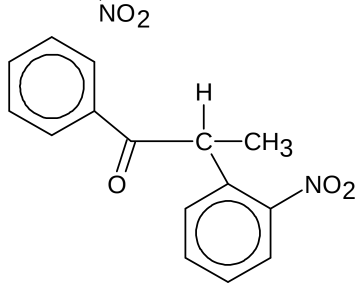
(D) Ph_3CCl

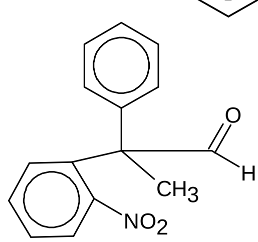
36. What is the reaction product of the following reaction?



(A) 

(B) 

(C) 

(D) 

37. $CHCl_3$ (Chloroform) is stored in closed dark coloured bottles and completely filled…….
(A) As it is decomposed to give CH_2Cl_2 and Cl_2 in presence of light and Cl_2 is poisonous.
(B) As it is decomposed to give (CCl_4 and HCl) in presence of light and HCl is poisonous and acidic as well.
(C) As it slowly oxidized by air in presence of light to an extremely poisonous gas, $COCl_2$ (Phosgene).
(D) As it is slowly oxidized to give CO and HCl, which are poisonous.

38. Which of the following is correct?
(A) If BOD level of water in reservoir is less than 10 ppm, it is highly polluted.
(B) High BOD means low activity of bacteria in water.
(C) Excessive use of chlorinated synthetic pesticides cause soil and water pollution.
(D) Synthetic chlorinated pesticides are biodegradable.

39. Which of the following is sulphur containing compound/drug?
(A) Renitidine
(B) Brompheniramine
(C) Phenelzine
(D) Veronal

40. Which types of isomerism is not exhibited by the compound with molecular formula $CrCl_2Br.6H_2O$?
(A) Hydrate isomerism
(B) Ionization isomerism
(C) Geometrical isomerism
(D) Optical isomerism

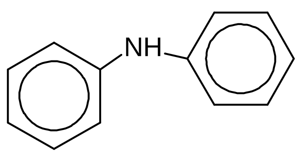
41. During the extraction of copper, Cu-metal is formed in Bessemer converter due to the reaction…..?
(A) $2Cu_2O \xrightarrow{C} 4Cu + CO_2;$
(B) $Cu_2S \xrightarrow{\Delta} 2Cu + S;$
(C) $Fe + Cu_2O \longrightarrow 2Cu + FeO;$
(D) $Cu_2S + 2Cu_2O \longrightarrow 6Cu + SO_2;$

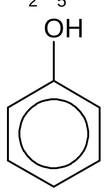
42. Which of the following is correct regarding gas 'P' evolved in the following process?
$Zn + dil.HNO_3 \longrightarrow Zn(NO_3)_2 + P\uparrow;$
(A) 'P' evolve as brown fumes.
(B) Aqueous solution of 'P' is acidic in nature.
(C) 'P' is a neutral gas.
(D) 'P' forms brown ring with $FeSO_4$ solution.

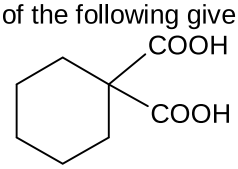
43. Which of the following is not soluble in aqueous NaOH?
(A) ZnO
(B) Fe_2O_3
(C) Al_2O_3
(D) BeO

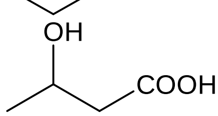
44. Which of the following is formed as white turbidity, on dissolving $BiCl_3$ in excess of water?
(A) $Bi(OH)_3$
(B) Bi_2O_3
(C) Bi_2OCl_3
(D) $BiOCl$

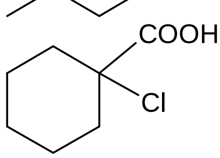
45. A white salt (P), turn black by reaction with NH_3, forming the compound (Q), which is soluble in aqueous aqua-regia and gives compound (R). 'R' gives orange red precipitate with KI which is soluble in excess of KI, forms (S). S is used to test $NH_3(g)$. What is 'P'?
(A) $PbCl_2$
(B) K_2HgI_4
(C) Hg_2Cl_2
(D) $HgCl_2$

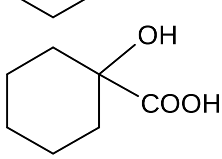
46. Which of the following give foul smell on treating with $KOH / CHCl_3$?
(A)


(B) $C_2H_5 - NH_2$
(C) $C_2H_5 - OH$
(D)


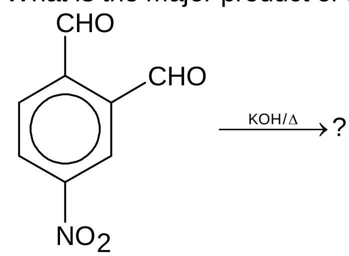
47. Which of the following give CO gas on heating with dil. Acid?
(A)


(B)


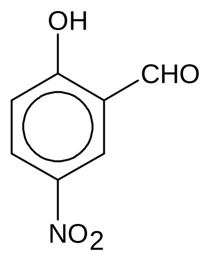
(C)


(D)


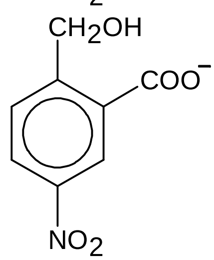
48. What is the major product of the following reaction?



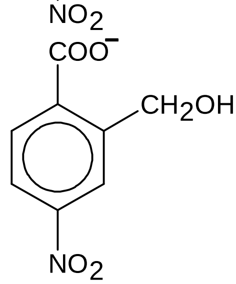
(A)



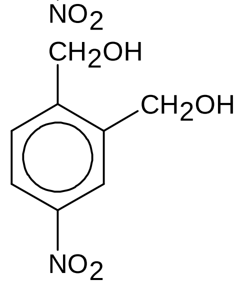
(B)



(C)



(D)



49. In a solid crystal of a mixed oxide, the oxide ions constitute ccp lattice. Element ‘A’ occupy half of the octahedral voids while the element ‘B’ occupy 12.5% of tetrahedral voids. The formula of the oxide is……..?

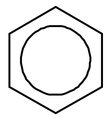
(A) $A_2B_3O_4$

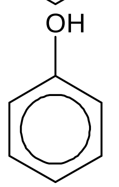
(B) AB_2O_3

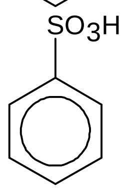
(C) A_2BO_4

(D) AB_2O_2

50. By which of the following process, benzene is produced as ultimate product?

(A)  $+ CH_3Cl \xrightarrow{\text{Anhyd. } AlCl_3} \xrightarrow{KMnO_4} \xrightarrow{H^+/\Delta} ?$

(B)  $\xrightarrow{\text{Zn dust}} ?$

(C)  $\xrightarrow{\text{Sunlight}} ?$

(D) By all the three processes.

SECTION – B
(Numerical Answer Type)

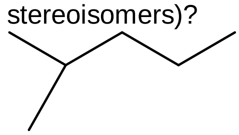
This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. How many 'N' atoms are there, per molecule of Adenine?

52. How many of the following are copolymers?
Buna-S, Teflon, Cellulose, Dacron, Nylon-6,6, Bakelite, Novolac, Neoprene, Buna-N, Glyptal

53. How many chiral centres are there per molecule of sucrose?

54. How many isomeric monochlorinated products of the following are possible (including stereoisomers)?

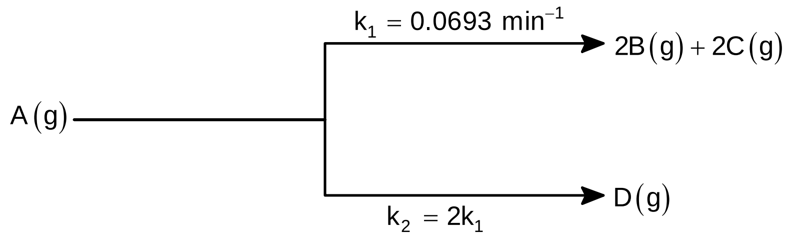


55. How many moles of sucrose should be dissolved in 500 gm of water to prepare a solution, which has a difference of 104.76°C between its melting point and boiling point?
$\left(K_f = 1.86 \text{ K / molal}; \; K_b = 0.52 \text{ K / molal}\right)$

56. Half-life of a zero order reaction: $R \rightarrow P$; is 6 min. How much total time (min) is required to complete the reaction by 75%?

57. 20 ml of a standard hydro gold sol requires 36 mg sodium oleate to protect the gold sol from coagulation by 3 ml 20% NaCl solution. What is the gold number of sodium oleate sample?

58.



Find partial pressure of 'A' (in atm) at 10 min, if initial pressure of A was 16 atm and none of B, C and D was present initially.

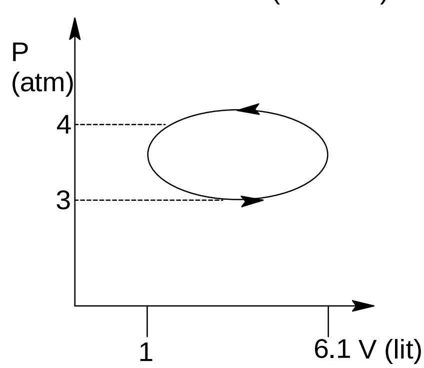
59. What is the value of K_{eq} for the following cell reaction?

$Cu(s) \mid CuNO_3(1M) \parallel I_2(s) \mid I^-(1\,M); Pt;$ at 25^oC.

If $E^o_{Cu^+|Cu} = +0.5196;$ $\quad E^o_{I_2|I^-} = 0.534$

[Given: log 3 = 0.48; $\frac{2.303RT}{F} = 0.06$]

60. What is the work done (in atm lit) during the following process?



Mathematics PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Let a_m be the m^{th} term of the sequence $\left(\frac{2^2}{1^2}-\frac{2}{1}\right)^{-1}, \left(\frac{3^3}{2^3}-\frac{3}{2}\right)^{-2}, \left(\frac{4^4}{3^4}-\frac{4}{3}\right)^{-3}$, ….., then the value of $\lim\limits_{m\to\infty}\left(a_m\right)^{\frac{1}{m}}$ is equal to
(A) $e + 1$
(B) $e - 1$
(C) $\frac{1}{e-1}$
(D) $\frac{1}{e+1}$

62. Let f be a polynomial function such that for all real x, $f(x^2 + 1) = x^4 + 5x^2 + 2$, then $\int f(x)dx$ is
(A) $\frac{x^3}{3}+\frac{3x^2}{2}-2x+c$
(B) $\frac{x^3}{3}+\frac{3x^2}{2}+2x+c$
(C) $\frac{x^3}{3}-\frac{3x^2}{2}-2x+c$
(D) $\frac{x^3}{3}-\frac{3x^2}{2}+2x+c$

63. Let $\alpha > -1$ and $\beta > -1$, then the value of $\lim\limits_{n\to\infty} n^{\beta-\alpha}\left(\frac{1^\alpha+2^\alpha+...+n^\alpha}{1^\beta+2^\beta+3^\beta+.....n^\beta}\right)$ is
(A) $\frac{\beta+1}{\alpha+1}$
(B) $\frac{\alpha+1}{\beta+1}$
(C) $\frac{\alpha+2}{\beta+2}$
(D) $\frac{\beta+2}{\alpha+2}$

64. A circle with centre at (15, –3) is tangent to $y=\frac{x^2}{3}$ at a point in the first quadrant. The radius of the circle is equal to
(A) $5\sqrt{6}$
(B) $8\sqrt{3}$
(C) $9\sqrt{2}$
(D) $6\sqrt{5}$

65. Let $f : R \to R$ be a function defined by $f(x) = 2x^3 - 21x^2 + 78x + 24$ the number of integers in the solution set of x satisfying the inequality $f(f(f(x) - 2x^3)) \geq f(f(2x^3 - f(x)))$ is
(A) 3
(B) 4
(C) 5
(D) 6

66. $x_1 = \sqrt[7]{3} + \sqrt[7]{9}$, $x_2 = \sqrt[7]{5} + \sqrt[7]{7}$, then
(A) $x_1 = x_2$
(B) $x_1 > x_2$
(C) $x_1 < x_2$
(D) none of these

67. If the system of equations $x + y = 1$, $(c + 2)x + (c + 4)y = 6$ and $(c + 2)^2x + (c + 4)^2y = 36$ are consistent and c_1, c_2 ($c_1 > c_2$) are two values of c, then c_1c_2 is divisible by
(A) 3
(B) 6
(C) 9
(D) 8

68. If $a = \log_{12} 18$ and $b = \log_{24} 54$, then the value of $(a + b)^2 + a(10 - a) - b(10 + b)$ is equal to
(A) $\frac{1}{2}$
(B) 1
(C) 2
(D) 3

69. If OABC is a tetrahedron such that $OA^2 + BC^2 = OB^2 + CA^2 = OC^2 + AB^2$, then which one of the following is INCORRECT?
(A) $OA \perp BC$
(B) $OB \perp CA$
(C) $OC \perp AB$
(D) $AB \perp BC$

70. If PSQ is a focal chord of the parabola $y^2 = 8x$ such that SP = 6, then the length of SQ is
(A) 6
(B) 4
(C) 3
(D) none of these

71. If the normal at any point P of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (a > b > 0) meets the major and minor axes at G ad G′ respectively and OF is the perpendicular drawn from centre O to this normal, then $PF \cdot PG$ is equal to
(A) b^2
(B) a^2
(C) ab
(D) none of these

72. If ABC is a triangle whose internal angles are A, B, C and ΔDEF is pedal triangle of ΔABC, then the ratio of the area of ΔDEF to area of ΔABC is
(A) 2 cos A cos B cos C
(B) 2 sin A sin B sin C
(C) 2 cos A cos B sin C
(D) none of these

73. In a triangle PQR let $\angle PQR = 30^\circ$ and the sides PQ and QR have lengths $10\sqrt{3}$ and 10 respectively, then which of the following statement is NOT TRUE?
(A) $\angle QPR = 45^\circ$
(B) the area of ΔPQR is $25\sqrt{3}$ and $\angle QPR = 120^\circ$
(C) the radius of the incircle of ΔPQR is $10\sqrt{3}-15$
(D) area of the circumcircle of ΔPQR is 100π

74. If $D = \text{diag}\left[d_1, d_2, d_2, ... d_n\right]$ is a diagonal matrix with $d_i \neq 0$ for i = 1, 2, 3, ….. n, then D^{-1} is
(A) $d_1^{-1}\, d_2^{-1}\, d_3^{-1}\, ...\, d_1^{-1}\, I_n$
(B) $\text{diag}\left[d_n^{-1}, d_{n-1}^{-1}, d_{n-2}^{-1}, ...\, d_1^{-1}\right]$
(C) $\text{diag}\left[d_1^{-1}, d_2^{-1}, d_3^{-1}, ...\, d_n^{-1}\right]$
(D) none of these

75. In ΔABC cos A + 2 cos B + cos C = 2, then the sides of the triangle are in
(A) AP
(B) GP
(C) HP
(D) none of these

76. Let PQ be a double ordinate of parabola $y^2 = 4ax$. Let N be the point of intersection of normals drawn at P and Q. Let T be intersection of tangents at P and Q. Then which one is NOT CORRECT
(A) area of Δ formed by circum-centres of ΔPQN, ΔPQT and ΔTQN is zero
(B) $\angle NQT = \frac{\pi}{2}$
(C) circle passing through P, T and N also passes through the point Q
(D) none of these

77. If $\int_0^{100} f(x)\,dx = a$, then $\sum_{r=1}^{100}\left(\int_0^1 f(r-1+x)\,dx\right)$ is equal to
(A) 0
(B) a
(C) $\frac{a}{2}$
(D) 2a

78. The common chord of the circle $x^2 + y^2 + 8x + 4y - 5 = 0$ and a circle passing through the origin and touching the line y = x passes through the fixed point
(A) $\left(\frac{5}{12}, \frac{5}{12}\right)$
(B) $\left(\frac{5}{12}, -\frac{5}{12}\right)$
(C) $\left(-\frac{5}{12}, \frac{5}{12}\right)$
(D) $\left(-\frac{5}{12}, -\frac{5}{12}\right)$

79. Given $p(x) = x^4 + ax^3 + bx^2 + cx + d$ such that $x = 0$ is the only real root of $p'(x) = 0$. If $p(-1) < p(1)$, then in the interval $[-1, 1]$
(A) $p(-1)$ is the minimum and $p(1)$ is maximum value of $p(x)$
(B) $p(-1)$ is not minimum but $p(1)$ is the maximum value of $p(x)$
(C) $p(-1)$ is the minimum and $p(1)$ is not the maximum value of $p(x)$
(D) neither $p(-1)$ is the minimum nor $p(1)$ is the maximum value of $p(x)$

80. If $\bar{a}, \bar{b}, \bar{c}$ are non-coplanar unit vectors such that $\bar{a} \times (\bar{b} \times \bar{c}) = \dfrac{\bar{b} + \bar{c}}{\sqrt{2}}$, then angle between $\bar{a}$ and $\bar{b}$ is
(A) $\dfrac{\pi}{4}$
(B) $\dfrac{\pi}{2}$
(C) π
(D) $\dfrac{3\pi}{4}$

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. If $f(x) = x^3 + 3x + 4$ and g is the inverse of f, then the value of $9\left|\dfrac{d}{dx}\left(\dfrac{g(x)}{g(g(x))}\right)\right|$ at $x = 4$ is equal to

82. Number of integral values of a for which the equation $\sin^4 x - 2\cos^2 x + a^2 = 0$ has a solution is

83. Let $a_1, a_2, a_3, \ldots, a_n$ be n, $(n \geq 10)$ natural numbers in A.P. such that the volume of parallelopiped formed by vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = a_2\hat{i} + a_3\hat{j} + a_1\hat{k}$ and $\vec{c} = a_3\hat{i} + a_1\hat{j} + a_2\hat{k}$ is 108, then a_4 is equal to (given $a_1 \leq 6$)

84. Number of solutions of the equation $\sin^8 \dfrac{\pi}{2}x - \cos^2 \dfrac{\pi}{2}x = x^2 - 4x + 5$ in the interval $[-2\pi, 2\pi]$ is

85. Let A and B are two independent events such that $P(A) + P(B) = 1$, $P\left(\dfrac{A}{B}\right) = \dfrac{2}{3}$ and $P(\bar{A} \cap B) = x$, then $54x$ is equal to

86. If the equation of a plane parallel to y-axis and containing the points (1, 0, 1) and (3, 2, –1) is $ax + by + cz = 2$, then the value of $|a + b + c|$ is

87. Let a line with the inclination angle 60° (with +ve x-axis) be drawn through the focus F of the parabola $y^2 = 8(x + 2)$. If the two intersection points of the line and the parabola are A and B and the perpendicular bisector of the chord AB intersects the x-axis at the point P, the length of the line segment PF is λ, then $\dfrac{3}{4}\lambda$ is equal to

88. Let z_1 and z_2 be two complex numbers such that $|z_2| = 1$ and $\frac{z_1 - 1}{z_1 + 1} = \left(\frac{z_2 - 1}{z_2 + 1}\right)^2$, then the value of $|z_1 - 1| + |z_1 + 1|$ is equal to

89. Gaurav writes letters to his five friends. The number of ways can the letters be placed in the envelopes so that at least two of them are in the wrong envelopes are λ, then $\lambda - 110$ is equal to

90. $\dfrac{\int_0^{\infty} x^9 e^{-x^2} dx}{\int_0^{\infty} x^7 e^{-x^2} dx}$ is equal to